

CUSTOMER BEHAVIOUR ANALYSIS QUERIES

PROBLEM STATEMENT

1. WHAT IS THE TOTAL REVENUE GENERATED BY MALE VS FEMALE CUSTOMERS?

```
SELECT gender,  
       SUM(purchase_amount) AS revenue  
FROM customer_shopping  
GROUP BY gender;
```

OUTPUT:

Result Grid			Filter Rows:
	gender	revenue	
▶	Male	157890	
	Female	75191	

2. WHICH CUSTOMERS USED A DISCOUNT BUT STILL SPENT MORE THAN THE AVERAGE AMOUNT?

```
SELECT customer_id,  
       purchase_amount  
FROM customer_shopping  
WHERE discount_applied = 'Yes'  
AND purchase_amount >= (  
    SELECT AVG(purchase_amount)  
    FROM customer_shopping  
);
```

OUTPUT:

Result Grid			Filter Rows:	Result Grid			Filter Rows:	Result Grid			Filter Rows:
	customer_id	purchase_amount			customer_id	purchase_amount			customer_id	purchase_amount	
▶	2	64			20	90			1662	86	
	3	73			22	62			1667	64	
	4	90			24	88			1671	73	
	7	85			29	94			1673	73	
	9	97			32	79			1674	62	
	12	68			33	67			1676	90	
	13	72			35	91					

3. WHICH ARE THE TOP 5 PRODUCTS WITH THE HIGHEST AVERAGE REVIEW RATINGS?

```
SELECT item_purchased,  
       ROUND(AVG(review_rating),2) AS Average_Product_Rating  
FROM customer_shopping  
GROUP BY item_purchased  
ORDER BY AVG(review_rating) DESC  
LIMIT 5;
```

OUTPUT:

Result Grid			Filter Rows:
	item_purchased	Average_Product_Rating	
▶	Gloves	3.86	
	Sandals	3.84	
	Boots	3.82	
	Hat	3.8	
	Skirt	3.78	

4. COMPARE THE AVERAGE PURCHASE AMOUNTS BETWEEN STANDARD AND EXPRESS SHIPPING?

```
SELECT shipping_type,  
       ROUND(AVG(purchase_amount),2) AS avg_purchase_amount  
FROM customer_shopping  
WHERE shipping_type IN ('Standard','Express')  
GROUP BY shipping_type;
```

OUTPUT:

Result Grid			Filter Rows:
	shipping_type	avg_purchase_amount	
▶	Express	60.48	
	Standard	58.46	

5. DO SUBSCRIBE CUSTOMER SPEND MORE?

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customers,  
       ROUND(AVG(purchase_amount),2) AS avg_spend,  
       ROUND(SUM(purchase_amount),2) AS total_revenue  
FROM customer_shopping  
GROUP BY subscription_status
```

ORDER BY total_revenue DESC;

OUTPUT:

	subscription_status	total_customers	avg_spend	total_revenue
▶	No	2847	59.87	170436
	Yes	1053	59.49	62645

6. WHICH 5 PRODUCTS HAVE HIGHEST PERCENTAGE OF PURCHASE WITH DISCOUNTS APPLIED?

```
SELECT item_purchased,
       ROUND(
         100.0 * SUM(
           CASE
             WHEN discount_applied = 'Yes' THEN 1
             ELSE 0
           END
         ) / COUNT(*),
       2) AS discount_rate
FROM customer_shopping
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

OUTPUT:

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7.SEGMENT CUSTOMERS INTO NEW, RETURNING AND LOYAL.

```
WITH customer_type AS (
  SELECT customer_id,
         previous_purchases,
         CASE
           WHEN previous_purchases = 1 THEN 'New'
```

```

        WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
        ELSE 'Loyal'
    END AS customer_segment
FROM customer_shopping
)
SELECT customer_segment,
       COUNT(*) AS Number_of_Customers
FROM customer_type
GROUP BY customer_segment;

```

OUTPUT:

	customer_segment	Number_of_Customers
►	Loyal	3116
	Returning	701
	New	83

8.WHAT ARE TOP 3 MOST PURCHASED PRODUCTS WITHIN EACH CATEGORY?

```

WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (
               PARTITION BY category
               ORDER BY COUNT(customer_id) DESC
           ) AS item_rank
    FROM customer_shopping
    GROUP BY category, item_purchased
)
SELECT item_rank,
       category,
       item_purchased,
       total_orders
FROM item_counts
WHERE item_rank <= 3;

```

OUTPUT:

Result Grid

Filter Rows:

Export:

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160

Result Grid

Filter Rows:

Export:

	item_rank	category	item_purchased	total_orders
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

9. ARE CUSTOMERS WHO ARE REPEAT BUYERS ALSO LIKELY TO SUBSCRIBE?

```
SELECT subscription_status,  
       COUNT(customer_id) AS repeat_buyers  
FROM customer_shopping  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

OUTPUT:

Result Grid	Filter Rows:
subscription_status	repeat_buyers
Yes	958
No	2518

10. WHAT IS THE REVENUE CONTRIBUTION OF EACH AGE GROUP?

```
SELECT age_group,  
       SUM(purchase_amount) AS total_revenue  
FROM customer_shopping  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

OUTPUT:

Result Grid	Filter Rows:
age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763